MATHS MANTHAN
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Subject: Mathematics | **CBSE CLASS 10 – PERIODIC TEST 09** | **Time:** 60 Minutes
Class: Class X (CBSE) | **Chapter 9: Some Applications of Trigonometry** | **MM: 30**

Student Name: _____ **Roll No.:** _____ **Date:** _____

GENERAL INSTRUCTIONS:

1. All questions are compulsory. Internal choice is provided in Case Study Question 12(iii).
2. **Section A** comprises 4 MCQs and 1 Assertion-Reason question of 1 mark each.
3. **Section B** comprises 2 Very Short Answer questions of 2 marks each.
4. **Section C** comprises 2 Short Answer questions of 3 marks each.
5. **Section D** comprises 2 Long Answer / HOTS questions of 5 marks each.
6. **Section E** comprises 1 Applied Case Study question of 5 marks with internal choice.
7. Use of calculators is not permitted. Take $\sqrt{3} = 1.732$ unless stated otherwise.

SECTION A: Multiple Choice & Assertion-Reason (5 × 1 = 5 Marks)

Q1. If the sun's elevation increases from 30° to 45°, the length of the shadow of a tower: [1]

(A) increases (B) decreases (C) remains identical (D) doubles

Q2. The angle of elevation of the top of a 15 m high tower from a point 15 m away from its foot on level ground is: [1]

(A) 30° (B) 45° (C) 60° (D) 90°

Q3. A ladder rests against a vertical wall at an angle α to horizontal. Its foot is pulled away by distance p and slides down distance q making angle β. Then $\frac{p}{q}$ equals: [1]

(A) $\frac{\cos\beta - \cos\alpha}{\sin\alpha - \sin\beta}$ (B) $\frac{\cos\alpha - \cos\beta}{\sin\beta - \sin\alpha}$ (C) $\frac{\sin\alpha - \sin\beta}{\cos\beta - \cos\alpha}$ (D) 1

Q4. From a point P on level ground, the angle of elevation of the top of a tower is 30°. If the tower is 100 m high, the distance of point P from the foot of the tower is: [1]

(A) $100\sqrt{3}$ m (B) $\frac{100}{\sqrt{3}}$ m (C) $50\sqrt{3}$ m (D) 200 m

Q5. **Assertion (A):** If a vertical pole 6 m high casts a shadow $6\sqrt{3}$ m long on the ground, then the sun's elevation is 60°.
Reason (R): For a vertical pole of height h and shadow s, $\tan\theta = \frac{h}{s}$; here $\tan\theta = \frac{6}{6\sqrt{3}} = \frac{1}{\sqrt{3}} \implies \theta = 30°$. [1]

(A) Both (A) and (R) are true; (R) explains (A).
(B) Both are true; (R) does not explain (A).
(C) (A) is true but (R) is false.
(D) (A) is false but (R) is true.

SECTION B: Very Short Answer Questions (2 × 2 = 4 Marks)

Q6. The tops of two poles of heights 18 m and 12 m are connected by a wire. If the wire makes an angle of 30° with the horizontal, find the length of the wire. [2]

Q7. An electrician has to repair an electric fault on a pole of height 5 m. She needs to reach a point 1.3 m below the top using a ladder inclined at 60° to the horizontal. What should be the length of the ladder? [Take $\sqrt{3} = 1.732$] [2]

SECTION C: Short Answer Questions (2 × 3 = 6 Marks)

Q8. From the top of a chimney 60 m high, the angles of depression of two points on the ground on the same side are observed to be 30° and 45°. Find the distance between the two points. [Take $\sqrt{3} = 1.732$] [3]

Q9. An aeroplane flying at a constant altitude of 2500 m is observed from a tracking station at an angle of elevation of 60°. After 20 seconds, the angle of elevation drops to 30°. Find the speed of the aeroplane in km/h. [3]

SECTION D: Long Answer / HOTS Questions (2 × 5 = 10 Marks)

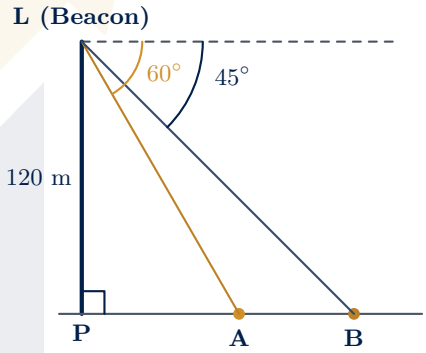
Q10. A round spherical balloon of radius r subtends an angle α at the eye of an observer while the angle of elevation of its centre is β. Prove that the height of the centre of the balloon above the horizontal observer plane is: [5]

$$h = r \sin\beta \csc\frac{\alpha}{2}$$

Q11. An observer standing on the deck of a ship, 10 m above water level, observes that the angle of elevation of the top of a hill is 60° and the angle of depression of the base of the hill is 30°. Calculate the distance of the hill from the ship and the total height of the hill. [5]

SECTION E: Applied Case Study (1 × 5 = 5 Marks)

Q12. CASE STUDY: Coastal Lighthouse Maritime Harbor Safety



A maritime navigation lighthouse is situated atop a coastal cliff. The luminous beacon lamp is located at point L, which is at an effective height of 120 m above sea level ($LP = 120$ m).
The lighthouse keeper observes two merchant vessels, A and B, approaching the harbor along the same radial line. The angle of depression of vessel A (closer) is 60°, while the angle of depression of vessel B (farther) is 45°.

Based on the above mathematical scenario, answer the following questions:

(i) Find the horizontal distance of vessel B from the base of the lighthouse (P). [1]

(ii) Find the horizontal distance of vessel A from the base of the lighthouse (P). [2]

(iii) Calculate the distance between vessel A and vessel B. [Take $\sqrt{3} = 1.732$] [2]

— OR —

If vessel A requires 2 minutes to cover the distance between vessel B and vessel A, find its average speed in m/s. [2]

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★★★ END OF QUESTION PAPER · ALL THE BEST ★★★